

CR-CA: A Two-Tier Framework for Structured Causal Reasoning and Counterfactual Analysis

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Abstract—CR-CA (Causal Reasoning – Counterfactual Analysis) is a two-tier framework for causal and counterfactual analysis. The strict tier (`crca_core`) provides formal causal identification, estimation, and spec lifecycle management under explicit assumptions. The heuristic tier (`CRCAAgent`) provides LLM-integrated causal exploration, automatic variable extraction, and deterministic causal simulation with learned edge coefficients and polynomial mechanism discovery. The framework’s core contribution is not to replace statistical causal inference or guarantee causal truth, but to make the boundary between formal and heuristic reasoning explicit, auditable, and configurable.

This paper reports the results of a comprehensive code audit and formal benchmark suite (v2.0.0) that validates the implementation against Pearl’s causal hierarchy. The audit reveals that the heuristic tier satisfies formal correctness requirements only when explicitly configured for linear mechanisms, acyclic SCMs, and descendant-only propagation. The main conclusion is that CR-CA is technically defensible when treated as a structured interface with clear epistemic boundaries, not as an assumption-free engine for deriving true causes from arbitrary text. Empirical benchmark results show that CR-CA’s heuristic tier outperforms naive OLS on collider scenarios (0.50x Oracle error) and achieves competitive results on chain (2.61x) and fork (5.69x) scenarios, but requires proper causal identification algorithms (backdoor/frontdoor/IV) to close the gap on confounded scenarios.

Github Repository: <https://github.com/Euroswarms-Institute/CR-CA>

Index Terms—causal reasoning, counterfactual analysis, structural causal models, Pearl’s causal hierarchy, LLM integration

1 INTRODUCTION

Modern language and agent systems often produce plausible chains of reasoning without a stable distinction between observation, mechanism, intervention, and counterfactual. Pearl’s structural causal model tradition formalizes causality through variables, structural equations, graphs, interventions, and counterfactuals; it also distinguishes causal questions from purely associational statistical questions.

CR-CA is motivated by this distinction. Unlike prior work that treats LLM-integrated causal reasoning as uniformly valid or uniformly invalid, CR-CA explicitly separates two tiers:

Tier 1 (`crca_core`): A strict causal inference package with `DraftSpec` $\rightarrow$ `LockedSpec` lifecycle, backdoor/frontdoor/IV/ID identification algorithms, and structured refusal under non-identifiability.

Tier 2 (`CRCAAgent`): An LLM-integrated agent with automatic OLS-learned edge coefficients, polynomial mechanism

discovery, and rich domain extensions (socioeconomic dynamics, corporate governance, quantitative trading).

This separation is the central architectural contribution. It prevents the heuristic tier from making unchecked formal claims, and it prevents the formal tier from being buried under usability features.

The core premise is that reasoning systems should explicitly separate at least six classes of information: observations, hypotheses, mechanisms, interventions, counterfactuals, and uncertainty. This separation does not guarantee truth. It makes errors more inspectable. A model can still infer the wrong graph, assume an invalid mechanism, or evaluate a counterfactual outside the support of available evidence. However, these failures become localized rather than hidden inside fluent prose.

In the strict interpretation, CR-CA is not a domain-specific trading system, not a visual world model, not a deployment product, and not a general AGI architecture. It is a framework whose job is to receive a problem representation, construct or accept a causal representation, perform intervention/counterfactual analysis, and return structured reasoning traces with uncertainty and consistency checks.

2 BACKGROUND

2.1 Association, Intervention, and Counterfactuals

Pearl’s causal hierarchy separates reasoning into three broad levels: association, intervention, and counterfactual reasoning. Associational questions concern passive observation, such as $P(Y|X = x)$. Interventional questions concern manipulated systems, such as $P(Y|do(X = x))$. Counterfactual questions concern alternative outcomes for a specific unit or world, such as “given that $X = x$ and $Y = y$ occurred, what would Y have been if X had instead been x' ?”

CR-CA’s formal tier belongs to all three levels. Its heuristic tier belongs primarily to levels two and three under configured constraints. Its core value is not in predicting correlations, but in forcing the reasoning process to ask whether a statement is observational, interventional, or counterfactual.

2.2 Structural Causal Models

A structural causal model may be written as:

$$\mathbf{M} = (\mathbf{U}, \mathbf{V}, \mathbf{F}, P(\mathbf{U})) \quad (1)$$

Tier 1 – crca_core (Formal): Spec lifecycle (DraftSpec $\rightarrow$ LockedSpec), explicit epistemic metadata on every edge, refusal-first under non-identifiability, backdoor/frontdoor/IV/ID algorithms.

Tier 2 – CRCAAgent (Heuristic): Automatic variable extraction, deterministic causal simulation, OLS edge coefficient estimation, polynomial mechanism discovery, counterfactual abduction-action-prediction.

Fig. 1: The two-tier architecture separates formal causal identification (crca_core) from heuristic exploration and simulation (CRCAAgent).

where $\mathbf{U}$ is a set of exogenous variables, $\mathbf{V}$ is a set of endogenous variables, $\mathbf{F}$ is a set of structural functions, and $P(\mathbf{U})$ is a distribution over exogenous noise or background factors. Each endogenous variable $V_i \in \mathbf{V}$ is determined by a function:

$$V_i := f_i(Pa_i, U_i) \quad (2)$$

where Pa_i are the parents of V_i in the causal graph. This graph is often represented as a directed acyclic graph when there are no feedback loops, although cyclic and dynamic systems require additional care.

CR-CA’s formal tier assumes acyclic SCMs by default. The heuristic tier supports acyclic SCMs in its linear mode; its default nonlinear mode (tanh-activated) is an engineering heuristic without formal justification in Pearl’s framework and should be treated as a variance-stabilizing approximation rather than a causal mechanism.

2.3 Potential Outcomes

The potential outcomes framework defines causal effects by comparing outcomes under different treatments for the same unit or comparable units. A simple treatment effect may be written as:

$$\tau_i = Y_i(1) - Y_i(0) \quad (3)$$

CR-CA’s formal tier is closer in spirit to structural causal models than to pure potential outcomes, because it explicitly emphasizes mechanisms, graph structure, interventions, and branching worlds. However, potential outcomes remain relevant when CR-CA must express treatment comparisons without committing to a full mechanistic graph.

3 ARCHITECTURE: THE TWO-TIER DESIGN

3.1 Tier 1: crca_core (Formal)

crca_core is a small, strict package for causal inference. Its design principles are:

- **Spec lifecycle:** DraftSpec $\rightarrow$ LockedSpec. Only LockedSpec may authorize numeric causal outputs.
- **Explicit assumptions:** Every graph, edge, and structural equation carries epistemic metadata.
- **Refusal-first:** When identification fails, the system returns a RefusalResult, not a guess.
- **Provenance:** Every identification result carries a ProvenanceManifest with a stable hash.

The core API:

```
identify_effect(locked_spec, treatment, outcome)
→ IdentificationResult | RefusalResult
```

Supported identification strategies:

- Backdoor criterion (with automatic adjustment set discovery)
- Frontdoor criterion (with mediator validation)
- Instrumental variables (with exclusion restriction checking)
- Conservative ID algorithm (for complex graphs with latent confounders)

3.2 Tier 2: CRCAAgent (Heuristic)

CRCAAgent is a lightweight causal reasoning agent with LLM integration. It implements:

- Automatic variable extraction from natural language via LLM tool calls
- Deterministic causal simulation using linear or nonlinear structural equations
- OLS-based edge coefficient estimation: learns causal edge strengths from observational data
- Polynomial mechanism discovery: discovers polynomial (nonlinear) terms in structural equations from data
- Counterfactual scenario generation with Monte Carlo uncertainty quantification
- Policy engine integration, image annotation, and Excel bridge (domain extensions)

CRCAAgent is NOT a formal causal discovery system. It is a structured reasoning interface that can be configured to respect formal constraints. Its deterministic simulation engine supports:

- Linear propagation (Pearl-compliant when model_class="acyclic_scm" and nonlinear_activation="identity")
- Polynomial propagation: discovers $f(v) = \dots + \beta \cdot \text{parent}^{\text{exp}} + \dots$ terms from data
- Nonlinear propagation (tanh-activated, heuristic only)
- Counterfactual abduction-action-prediction (Pearl Level 3)
- Nested counterfactuals

The OLS edge coefficient estimation fits $z_v = \sum_u \beta_u \cdot z_u$ (zero-intercept, by z-standardization) in z-standardized space for each node. This provides correct marginal causal effects when the graph structure is correct but individual edge coefficients are unknown. For chains ($X \rightarrow Y \rightarrow Z$), OLS on $Z \sim Y$ recovers the marginal effect β_{YZ} , which when combined with β_{XY} via the chain rule gives the correct causal effect $\beta_{XY} \cdot \beta_{YZ}$ even when individual structural coefficients are not separately identifiable. For confounded graphs, OLS edge coefficients do NOT give correct causal

effects – proper backdoor/frontdoor/IV adjustment is required.

3.3 Relationship Between Tiers

The correct relationship is hierarchical:

- For research and audit: Use `crca_core` directly. Lock specs, identify effects, estimate with do-calculus.
- For exploration and prototyping: Use `CRCAAgent`. Extract variables, simulate interventions, generate scenarios, discover polynomial terms.
- For production causal claims: `CRCAAgent` should delegate to `crca_core`. The agent constructs a `DraftSpec`, locks it, and calls `identify_effect` before returning numeric causal outputs.

This bridge is partially implemented but not enforced. Future work should make it mandatory.

4 MATHEMATICAL CORE

4.1 Baseline World

Let a world be represented as:

$$W = (\mathbf{M}, \mathbf{v}, \mathbf{u}, E) \quad (4)$$

where $\mathbf{M}$ is the structural model, $\mathbf{v}$ is an assignment of endogenous variables, $\mathbf{u}$ is an assignment of exogenous variables, and E is observed evidence.

The baseline world W is the model-conditioned reconstruction of the world that best explains the evidence. This corresponds loosely to the abduction step in Pearl-style counterfactual reasoning: infer background conditions compatible with the observed evidence before applying the counterfactual intervention.

4.2 Intervention

An atomic intervention $do(X = x')$ modifies the model by replacing the original structural equation for X with a constant assignment:

$$X := x' \quad (5)$$

The intervened model is denoted $\mathbf{M}_{do(X=x')}$ or $\mathbf{M}_{x'}$.

The crucial constraint is that intervention is not conditioning. CR-CA must not silently replace $P(Y|do(X = x))$ with $P(Y|X = x)$ unless an identification argument justifies doing so.

4.3 Counterfactual World

Given evidence $E = e$, a counterfactual query may be written as:

$$P(Y_{x'} = y \mid E = e) \quad (6)$$

In CR-CA terms, this means: (1) infer compatible background state $\mathbf{u}$ from $E = e$; (2) modify the model using $do(X = x')$; (3) propagate the same inferred background state through the modified model; (4) compare Y in the baseline and intervened worlds.

The baseline and counterfactual worlds should share the same exogenous background unless the intervention explicitly changes it.

4.4 Descendant-Only Propagation

A strict CR-CA rule (enforced in v1.5.1) is:

After intervening on X , only X , its descendants, and variables mechanistically affected by the intervention may change. Non-descendants remain fixed unless the intervention changes the model boundary or exogenous state.

Let $De(X)$ denote descendants of X in graph G . For a surgical intervention $do(X = x')$, CR-CA enforces:

$$V_j' = V_j \quad \text{for all } V_j \notin \{X\} \cup De(X) \quad (7)$$

This rule is valid for standard surgical interventions in acyclic SCMs. It is not automatically valid for cyclic systems, equilibrium systems, hidden common causes, or interventions that alter mechanisms globally. Therefore, CR-CA must mark this rule as conditional on the model class.

4.5 OLS Edge Coefficient Estimation

When the causal graph structure (edges) is known but edge coefficients are unknown, CR-CA can estimate them from observational data using OLS regression in z-standardized space.

For a node V_j with parents Pa_j , the structural equation in standardized space is:

$$z_j = \sum_{u \in Pa_j} \beta_{uj} \cdot z_u + \varepsilon_j \quad (8)$$

where $z_j = (V_j - \mu_j)/\sigma_j$ is the standardized variable and $\varepsilon_j \sim \mathcal{N}(0, \sigma_\varepsilon^2)$. CR-CA fits this via OLS and uses the estimated β_{uj} as the edge strength for causal propagation. Note that because both z_j and each z_u have mean zero by construction, the intercept term vanishes and is omitted from the reported equation – the code includes an intercept column but the fitted value is effectively zero for z-standardized variables.

This approach is correct when:

- The graph structure (edges) is correctly specified
- The structural equations are linear in z-space (possibly with polynomial terms)
- There is no hidden confounding (all confounders are observed and in the graph)

This approach has fundamental limitations:

- In the presence of confounders, OLS edge coefficients are not structural coefficients and do not give correct causal effects. For example, in a fork $Z \rightarrow X, Z \rightarrow Y, X \rightarrow Y$ with confounder Z , the OLS coefficient of X on Y reflects the confounding effect of Z , not the causal effect of X on Y .
- For chains $X \rightarrow Y \rightarrow Z$, individual edge coefficients (β_{XY}, β_{YZ}) are not identifiable from observational data – only their product $\beta_{XY} \cdot \beta_{YZ}$ is. OLS on $Z \sim Y$ gives the correct marginal effect for predicting Z from Y , which yields the correct causal effect for $do(X)$ on Z via the chain rule.

4.6 Polynomial Mechanism Discovery

CR-CA can discover polynomial (nonlinear) terms in structural equations from data. For each candidate edge $u \rightarrow v$, it tests whether adding u^{exp} (for $\text{exp} \in \{2, 3\}$) to the regression of v on u significantly improves fit, using an incremental R^2 threshold (default 0.01).

The procedure:

- 1) Standardize: $z_u = (u - \mu_u)/\sigma_u$, $z_v = (v - \mu_v)/\sigma_v$
- 2) Baseline: Fit $z_v = \beta_0 + \beta_1 \cdot z_u + \varepsilon$. Record per-capita residual sum of squares $\hat{\sigma}_0^2 = \text{RSS}_0/n$.
- 3) For $\text{exp} \in \{2, 3\}$: Fit $z_v = \beta_0 + \beta_1 \cdot z_u + \beta_{\text{exp}} \cdot z_u^{\text{exp}} + \varepsilon$. Record $\hat{\sigma}_{\text{exp}}^2 = \text{RSS}_{\text{exp}}/n$.
- 4) If $\hat{\sigma}_0^2 - \hat{\sigma}_{\text{exp}}^2 \geq r^2_threshold$, accept the polynomial term.

The threshold $r^2_threshold = 0.01$ is a per-unit variance improvement threshold. Equivalently: $\text{RSS}_0 - \text{RSS}_{\text{exp}} \geq r^2_threshold \cdot n$.

When polynomial terms are discovered, they are added to the structural equation alongside linear terms. The linear term may be disabled via `disable_linear_for_polynomial()` to avoid double-counting when the relationship is purely polynomial.

4.7 Decision Scoring

CR-CA may support decision optimization, but only after causal states are represented. A generic decision score can be written as:

$$\text{Score}(a) = \mathbb{E}[U(W_a)] - \lambda R(W_a) - \gamma C(a) \quad (9)$$

where a is an action or intervention, $W_a = (\mathbf{M}_{do(a)}, \mathbf{v}_{do(a)}, \mathbf{u}, E)$ is the world resulting from applying intervention $do(a)$ to baseline world W , U is utility, R is risk, C is cost, and λ, γ are weighting parameters.

This scoring expression is not a theorem. It is an engineering interface. The utility, risk, and cost terms must be domain-defined. CR-CA should not hide normative choices inside mathematical notation.

5 PROPOSED ARCHITECTURE

5.1 Microkernel Interpretation

The cleanest architecture for the formal tier is a microkernel architecture:

$$K = (V, E, F, do, cf, score, audit) \quad (10)$$

where:

- V manages variables.
- E manages evidence.
- F manages mechanisms.
- do applies interventions.
- cf evaluates counterfactuals.
- $score$ ranks candidate worlds or actions.
- $audit$ records assumptions, uncertainty, and consistency violations.

The kernel should remain small. Domain adapters, language interfaces, memory systems, LLM calls, simulators, UI layers, and datasets should live outside the core.

5.2 Internal Pipeline

A disciplined CR-CA pipeline is:

Input $\rightarrow$ Parse $\rightarrow$ Boundary $\rightarrow$ Variables $\rightarrow$ Graph $\rightarrow$ Mechanism
 $\rightarrow$ Evidence $\rightarrow$ Abduction $\rightarrow$ Intervention $\rightarrow$ Propagation
 $\rightarrow$ Comparison $\rightarrow$ Audit

Each stage should expose failure states. For example:

- insufficient variables;
- ambiguous graph;
- no identified causal effect;
- intervention outside model support;
- ungrounded mechanism;
- counterfactual depends on hidden exogenous assumptions;
- utility function underspecified.

A mature CR-CA implementation should prefer returning “not identifiable under current assumptions” over producing a plausible but unjustified answer.

6 FORMAL CORRECTNESS REQUIREMENTS

A mathematically disciplined CR-CA core should satisfy the following requirements.

6.1 Conditioning and Intervention Separation

CR-CA must distinguish $P(Y|X = x)$ from $P(Y|do(X = x))$. These expressions are equal only under specific conditions. Treating them as generally equivalent is a causal error.

6.2 Explicit Model Class

CR-CA must state whether it assumes acyclic SCM, cyclic SCM, temporal dynamic model, equilibrium model, potential-outcomes-only model, or simulator-defined transition model. The descendant-only propagation rule is cleanest under acyclic SCM assumptions. It must not be blindly applied to dynamic feedback systems.

6.3 Mechanism Preservation Under Intervention

For a surgical intervention, CR-CA must preserve all non-intervened mechanisms:

$$f'_i = f_i \quad \text{for all } i \neq X \quad (11)$$

unless the intervention is explicitly a mechanism-changing intervention.

6.4 Counterfactual Same-Unit Consistency

A counterfactual world should preserve the inferred exogenous background state unless explicitly changed: $\mathbf{u}' = \mathbf{u}$. This is what makes the counterfactual about the same case rather than a new sampled case.

6.5 Non-Identifiability Handling

When a causal query is not identifiable from the supplied graph and data, CR-CA must return a non-identifiability result rather than silently estimating an associational proxy. This aligns with the logic of do-calculus and causal identification theory.

CR-CA satisfies 4.5 of 5 formal requirements when configured correctly:

- **Requirement 1 – Conditioning/Intervention separation:** PASS. The do-operator is implemented; $P(Y|X) \neq P(Y|do(X))$ is enforced.
- **Requirement 2 – Explicit model class:** PASS. `model_class="acyclic_scm"` is declared.
- **Requirement 3 – Mechanism preservation:** PASS. Descendant-only propagation is enforced.
- **Requirement 4 – Counterfactual same-unit consistency:** PASS. Abduction-action-prediction preserves the inferred exogenous background state.
- **Requirement 5 – Non-identifiability handling:** PARTIAL. The infrastructure exists but automatic abstention is disabled by default.

Overall: 4.5/5 requirements satisfied (90%).

7 EVALUATION: BENCHMARK RESULTS

CR-CA should be evaluated as a reasoning framework, not as a general chatbot. We conducted a formal benchmark suite covering Pearl’s causal hierarchy, intervention semantics, counterfactual consistency, confounding awareness, non-identifiability handling, propagation correctness, mechanism preservation, graph validity, and formal core behavior.

7.1 Benchmark Design

The expanded benchmark suite (`benchmark_expanded.py`) evaluates CRCA against Oracle (true SCM) and OLS Naive on 8 synthetic scenarios with known ground truth:

- **Chain** ($X \rightarrow Y \rightarrow Z$): Effect of $do(X)$ on Z through serial mediation.
- **Collider** ($X \rightarrow Z \leftarrow Y$): Effect of $do(X)$ on Z where Y is a common effect.
- **Fork** ($Z \rightarrow X, Z \rightarrow Y, X \rightarrow Y$): Effect of $do(X)$ on Y with confounder Z .
- **Multi-confounder** ($Z_1 \rightarrow X, Z_2 \rightarrow X, Z_1 \rightarrow Y, Z_2 \rightarrow Y, X \rightarrow Y$): Effect of $do(X)$ on Y with two confounders.
- **Frontdoor** ($X \rightarrow M \rightarrow Y, U \rightarrow X, U \rightarrow Y$): Effect of $do(X)$ on Y through mediator M with hidden confounder U .
- **Instrumental Variable** ($Z \rightarrow X \rightarrow Y, U \rightarrow X, U \rightarrow Y$): Effect of $do(X)$ on Y with valid instrument Z and hidden confounder U .
- **Nonlinear** ($X \rightarrow Y$ with quadratic nonlinearity $Y = 0.5 \cdot X^2 + \varepsilon$): Effect of $do(X)$ on Y with nonlinear mechanism.
- **Hidden Confounding** ($X \leftarrow U \rightarrow Y$): Effect of $do(X)$ on Y with unobserved confounder U .

Models tested: Oracle (true SCM), OLS Naive (regression of Y on X without adjustment), CR-CA Linear (heuristic agent with OLS edge estimation and polynomial discovery).

7.2 Intervention MAE Results

7.3 Counterfactual MAE Results

CRCA’s counterfactual performance is notably stronger on chain (0.0197 vs 0.4656), collider (0.0000 vs 0.7705), and fork (0.4885 vs 2.2571), demonstrating that its counterfactual abduction-action-prediction mechanism preserves same-unit consistency when the graph is correctly specified.

7.4 Key Findings

- 1) CRCA Linear outperforms OLS on collider ($0.50x < 0.56x$) and fork ($5.69x < 25.14x$) scenarios, demonstrating that descendant-only propagation provides correct intervention semantics when confounders are absent.
- 2) OLS is competitive on chain ($2.25x$ vs $2.61x$) because the marginal effect computed by OLS happens to equal the causal effect for serial mediation with unknown coefficients.
- 3) CRCA significantly underperforms OLS on multi-confounder ($32.49x$ vs $13.95x$) because OLS edge estimation cannot distinguish causal pathways through multiple confounders; this is not a CRCA bug but a fundamental limitation of edge-wise coefficient estimation without backdoor adjustment.
- 4) Frontdoor, hidden confounding, and IV scenarios remain unresolvable ($14\text{--}26x$ error) without implementing the corresponding causal identification algorithms (frontdoor criterion, IV/2SLS, or non-identifiable abstention).
- 5) Nonlinear scenarios fail ($37x$) because `crca_linear` uses identity activation without discovering the tanh nonlinearity; the polynomial discovery mechanism exists but is not yet integrated into `crca_nonlinear` mode for this benchmark.

8 SELF-CRITICAL ANALYSIS

8.1 What CR-CA Can Legitimately Claim

CR-CA can legitimately claim to provide a structured reasoning protocol for causal and counterfactual analysis. It can force a system to distinguish observation from intervention, hypothesis from mechanism, and baseline world from counterfactual world. It can improve auditability by making causal assumptions explicit.

The formal tier (`crca_core`) can claim:

- Backdoor/frontdoor/IV identification under explicit graph assumptions
- Structured refusal when identification fails
- Spec lifecycle with cryptographic hashing and approval tracking

The heuristic tier (CRCAAgent) can claim:

- Automatic variable extraction from natural language

TABLE 1: Intervention MAE Ratio vs. Oracle (lower is better; 1.0 = Oracle)

Scenario	CRCA Linear	OLS Naive	Notes
Collider	0.50x	0.56x	CRCA correctly avoids conditioning on descendant
Chain	2.61x	2.25x	OLS slightly better; OLS marginal effect works for serial mediation
Fork	5.69x	25.14x	CRCA 4.4x better; backdoor adjustment needed
Frontdoor	14.91x	16.39x	Both fail; frontdoor criterion not implemented
Hidden confound.	19.29x	18.83x	Both fail; hidden confounding not adjustable
IV	26.16x	24.14x	Both fail; IV/2SLS not implemented
Multi-confound.	32.49x	13.95x	CRCA worse; OLS edge estimation misleading with confounders
Nonlinear	37.05x	37.41x	Both fail; polynomial discovery not integrated in nonlinear mode

TABLE 2: Counterfactual MAE vs. Oracle (lower is better; 0.0 = Oracle)

Scenario	CRCA Linear	OLS Naive	Oracle
Chain	0.0197	0.4656	0.0000
Collider	0.0000	0.7705	0.0000
Fork	0.4885	2.2571	0.0000
Frontdoor	2.5940	3.4576	0.0000
Hidden confound.	3.4944	3.6779	0.0000
IV	1.3941	1.8966	0.0000
Multi-confound.	2.9392	2.1226	0.0000
Nonlinear	0.5693	0.5905	0.0000

TABLE 3: Counterfactual Coverage @0.5 (fraction of CF predictions within 0.5 of ground truth)

Scenario	CRCA Linear	OLS Naive
Chain	1.00	0.61
Collider	1.00	0.46
Fork	0.56	0.12
IV	0.22	0.13
Nonlinear	0.49	0.46
Frontdoor	0.09	0.06
Hidden confound.	0.09	0.12
Multi-confound.	0.11	0.15

- OLS-based edge coefficient estimation from observational data
- Polynomial mechanism discovery from data
- Fast deterministic simulation for prototyping
- Counterfactual scenario generation with uncertainty metadata
- Correct intervention semantics (descendant-only propagation) when graph is specified

8.2 What CR-CA Cannot Claim

CR-CA cannot claim that a generated graph is true merely because it is explicit. It cannot infer causal direction from text alone unless domain assumptions or data justify the inference. It cannot identify $P(Y|do(X = x))$ from $P(Y|X = x)$ without valid causal assumptions. It cannot resolve hidden confounding by formatting the problem better. It cannot make counterfactuals meaningful when the underlying mechanisms are absent or fictional. It cannot reliably beat OLS on all scenarios – the OLS edge coefficient estimation approach is fundamentally limited for confounded graphs.

It also cannot rely on chain-of-thought traces as proof of faithful reasoning. Research on chain-of-thought faithfulness shows that verbalized reasoning may not accurately

reveal the causal factors behind model outputs, and interventions on generated reasoning can expose unfaithfulness. Therefore, CR-CA should treat traces as audit artifacts, not as direct evidence that the underlying model actually reasoned correctly.

8.3 The Main Technical Risks

Risk 1 – Formal theater without causal identification: The system emits variables, graphs, equations, and counterfactual branches, but edge coefficient estimation does not substitute for proper causal identification algorithms. For scenarios with confounders (fork, multi_confounder, frontdoor, iv, hidden_confounding), the OLS-based edge strength approach produces wrong causal effects because OLS coefficients are associational, not causal.

Risk 2 – Polynomial discovery is not yet integrated into nonlinear mode: The polynomial discovery mechanism exists and works (improved nonlinear scenario from 158x to 37x Oracle error), but it is not yet integrated into the `crca_nonlinear` benchmark configuration. Full integration is future work.

Risk 3 – Abstention is not automatic: The `check_identifiability()` and

`abstain_on_nonidentifiable` flag exist, but automatic abstention under non-identifiability is disabled by default. Users must explicitly enable it.

Risk 4 – Two-tier gap: The formal tier (`crca_core`) with backdoor/frontdoor/IV identification is not yet wired into CRCAAgent. Users must manually check identifiability before trusting causal estimates.

8.4 The Nonlinearity Problem

The default nonlinear mode (nonlinear_activation="tanh") applies:

$$z_{\text{out}} = \tanh(z_{\text{in}}) \times 3.0 \quad (12)$$

This is an engineering heuristic, not a causal mechanism. It:

- Has no basis in Pearl’s structural equations
- Breaks exact counterfactual consistency (non-descendants drift)
- Is not invertible
- Is not documented in any causal inference textbook

The polynomial mechanism discovery provides a theoretically grounded alternative: discovering $f(v) = \beta \cdot \text{parent}^{\text{exp}} + \dots$ terms from data. This approach is valid when the structural equations contain polynomial terms (e.g., quadratic relationships) and the graph is correctly specified.

We recommend that users who need formal correctness set `use_nonlinear_scm=False` and `nonlinear_activation="identity"`. The nonlinear mode should be treated as a variance-stabilizing approximation for exploration only.

9 RELATIONSHIP TO LLMs

CR-CA can be implemented above an LLM as a structured reasoning layer. In that configuration, the LLM may perform parsing, hypothesis generation, explanation, or natural-language synthesis. The CR-CA core should perform constraint checking, intervention semantics, trace structuring, and counterfactual comparison.

However, the LLM should not be treated as an oracle for causal truth. LLMs can generate convincing causal arguments, but controlled counterfactual reasoning remains difficult, and chain-of-thought explanations can be unfaithful.

A strict architecture separates responsibilities as follows:

- LLM = linguistic interface and proposal generator
- CRCA core = causal structure, intervention, counterfactual, audit
- Data/Simulator = empirical grounding
- Planner = action selection

This separation prevents CR-CA from becoming a decorative prompt wrapper.

10 MINIMAL EXAMPLE

Suppose the system observes:

- $Z = \text{rain}$
- $X = \text{sprinkler active}$
- $Y = \text{wet grass}$

A naive model may infer: $X \rightarrow Y$

But a better causal model may include: $Z \rightarrow Y, Z \rightarrow X, X \rightarrow Y$

because rain can wet grass directly and may also influence whether a sprinkler is used.

A CR-CA query asks: “If the sprinkler had not been active, would the grass still have been wet?”

This is not simply $P(Y|X = 0)$. It is $P(Y_{X=0} | Z = \text{rain}, X = 1, Y = 1)$.

CR-CA should preserve the background condition $Z = \text{rain}$, intervene on X , and recompute Y . If rain alone is sufficient to wet the grass, then the counterfactual answer may remain $Y = 1$. If sprinkler was the only active cause, then Y may become 0. The result depends on the mechanisms, not only the graph.

This illustrates the central point: CR-CA is useful only if mechanisms are represented or at least honestly approximated.

11 LIMITATIONS

CR-CA’s first limitation is **causal graph dependence**: if the graph is wrong, the intervention and counterfactual outputs may be wrong even if the internal reasoning is formally consistent.

The second limitation is **mechanism underspecification**: graphs alone rarely determine exact counterfactual outcomes. Two models can share the same graph but imply different counterfactuals because their structural functions differ.

The third limitation is **hidden confounding**: if relevant exogenous variables are omitted, CR-CA may produce structurally clean but causally invalid conclusions. The OLS edge coefficient estimation approach is particularly vulnerable to this because OLS coefficients are biased under hidden confounding.

The fourth limitation is **LLM interface risk**: if an LLM proposes variables or mechanisms, CR-CA must treat them as hypotheses unless externally grounded.

The fifth limitation is **false precision**: mathematical notation should not be used where assumptions are too weak. A responsible CR-CA implementation should mark weak claims as weak.

The sixth limitation is **world-model dependence**: CR-CA can compare worlds only to the extent that the world representation exists. It is not itself a complete world model.

The seventh limitation is **the two-tier gap**: the formal tier and heuristic tier are not yet fully integrated. Users of CRCAAgent may not realize that their causal claims have not been validated by `crca_core`.

The eighth limitation is **polynomial discovery integration**: the polynomial mechanism discovery system is implemented but not yet integrated into the `crca_nonlinear` benchmark configuration, leaving nonlinear scenarios at high error.

12 CORE THESIS

The defensible thesis is:

CR-CA is a two-tier causal-reasoning framework. Its formal tier structures observations, mechanisms, interventions, and counterfactual worlds into an auditable reasoning process with explicit assumptions, valid causal graphs, and refusal under non-identifiability. Its heuristic tier provides LLM-integrated causal exploration, OLS-based edge coefficient estimation, polynomial mechanism discovery, and deterministic simulation. The framework’s value lies in enforcing causal discipline and making the boundary between formal and heuristic reasoning explicit. Its correctness depends on configuration, explicit assumptions, valid causal graphs, mechanism grounding, proper separation between observation, intervention, and counterfactual analysis, and – for confounded scenarios – proper causal identification algorithms (backdoor/frontdoor/IV). It should be evaluated not by fluency or apparent intelligence, but by intervention semantics, counterfactual consistency, uncertainty handling, and abstention under non-identifiability.

This thesis is intentionally modest. A stronger thesis would require empirical evidence that CR-CA improves causal reasoning accuracy across controlled benchmarks with proper causal identification algorithms, not merely that it produces better-looking reasoning traces or correct counterfactuals on simple graphs.

13 CONCLUSION

CR-CA is most mathematically coherent when defined as a framework with two tiers: a strict formal core for causal identification and estimation, and a heuristic agent for exploration, edge coefficient estimation, and simulation. The core should consist of variable representation, mechanism representation, intervention semantics, counterfactual world comparison, scoring, and audibility. It should not claim authority beyond its assumptions. It should not confuse graph generation with causal discovery, conditioning with intervention, or reasoning traces with faithful cognition.

The strongest version of CR-CA is therefore not the broadest one. It is the version that preserves causal correctness by making the formal-heuristic boundary explicit. Its purpose is to make causal reasoning explicit enough to be checked, rejected, refined, or grounded in data. That is a meaningful contribution, but only if the implementation remains disciplined and the paper accurately reflects what the code does.

The empirical results show that CR-CA’s heuristic tier already outperforms naive OLS on collider scenarios (0.50x Oracle error vs 0.56x) and fork scenarios (5.69x vs 25.14x), demonstrating that descendant-only propagation and OLS edge coefficient estimation provide meaningful improvements over standard regression when confounders are absent. However, CR-CA cannot close the gap on confounded scenarios (frontdoor, iv, hidden_confounding, multi_confounder) without implementing the corresponding causal identification algorithms from its formal tier. The path forward is integration: wiring `crca_core`’s backdoor/frontdoor/IV identification into `CRCAAgent`’s prediction pipeline.

14 AUDIT TRANSPARENCY STATEMENT

This paper was revised following a comprehensive code audit conducted on 2026-05-27. The audit found that the original implementation (v1.5.0) violated several formal requirements stated in the paper. Version 1.5.1 addresses these issues through:

- Descendant-only propagation enforcement
- Explicit model class declarations
- Assumption ledger with epistemic metadata
- Identifiability checking infrastructure
- Counterfactual non-descendant preservation
- OLS edge coefficient estimation from observational data
- Polynomial mechanism discovery from data
- Documentation of nonlinear heuristic limitations

The expanded benchmark suite (`benchmark_expanded.py`) and validation tests provide reproducible evidence. All results are available in the repository.

APPENDIX A: BENCHMARK CONFIGURATION

Benchmark configuration: $n = 5000$ train, $n = 100$ test, seed=42. All scenarios use linear SCMs unless otherwise noted. Noise $\varepsilon \sim \mathcal{N}(0, 1)$ unless noted.

- **Chain:** $X = \varepsilon, Y = 2.0 \cdot X + \varepsilon, Z = 0.5 \cdot Y + \varepsilon$
- **Collider:** $X = \varepsilon, Y = \varepsilon, Z = 1.0 \cdot X + 1.0 \cdot Y + \varepsilon$
- **Fork:** $Z = \varepsilon, X = 1.5 \cdot Z + \varepsilon, Y = 2.0 \cdot Z + 1.0 \cdot X + \varepsilon$
- **Multi-confounder:** $Z_1 = \varepsilon, Z_2 = \varepsilon, X = 0.3 \cdot Z_1 + 0.3 \cdot Z_2 + \varepsilon, Y = 0.5 \cdot X + 0.3 \cdot Z_1 + 0.3 \cdot Z_2 + \varepsilon$
- **Frontdoor:** $U = \varepsilon, X = 2.0 \cdot U + \varepsilon, M = 1.5 \cdot X + \varepsilon, Y = 1.0 \cdot M + 3.0 \cdot U + \varepsilon$
- **IV:** $Z = \varepsilon, U = \varepsilon, X = 1.0 \cdot Z + 2.0 \cdot U + \varepsilon, Y = 3.0 \cdot X + 1.5 \cdot U + \varepsilon$ (valid instrument Z ; unobserved U confounds X - Y)
- **Nonlinear:** $X = \varepsilon, Y = 0.5 \cdot X^2 + \varepsilon$ (quadratic, not tanh; noise $\varepsilon \sim \mathcal{N}(0, 0.5)$)
- **Hidden confounding:** $U = \varepsilon, X = 2.0 \cdot U + \varepsilon, Y = 1.0 \cdot X + 3.0 \cdot U + \varepsilon$

Intervention target: $do(X = X + 1)$ for all scenarios. Oracle evaluates the true SCM directly.

APPENDIX B: CONFIGURATION FOR FORMAL CORRECTNESS

To use CRCAAgent in a formally correct mode:

```
from CRCA import CRCAAgent
```

```
agent = CRCAAgent()
agent.set_model_class("acyclic_scm")
agent.use_nonlinear_scm = False
agent.nonlinear_activation = "identity"
agent.abstain_on_nonidentifiable = True
agent._assumption_ledger_enabled = True
```

All edges should be added with explicit epistemic status:

```
agent.add_causal_relationship(
    "X", "Y",
    strength=0.5,
    epistemic_status="supplied_by_user"
)
```

Before making causal claims, check identifiability:

```
result = agent.check_identifiability("X", "Y")
if not result["identifiable"]:
    raise ValueError(f"Not identifiable: {result['reason']}")
```

To use polynomial mechanism discovery:

```
agent.set_polynomial_terms_from_data(
    target="Y",
    parent="X",
    data_x=data["X"],
    data_y=data["Y"],
    max_exponent=3,
    r2_threshold=0.01
)
```

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